

✓ TF - IDF = Term frequency - inverse document frequency

Rečenice s kojima radimo:

S1: good boy

S2: good girl

S3: boy good girl

✓ TF - IDF se sastoji od dva dijela

TF = broj pojavljivanja određene riječi u rečenici / ukupan broj riječi

IDF = $\log(e)$ (ukupan broj rečenica / broj rečenica koje sadrže određenu riječ)

korištenjem ove tehnike hvata se važnost riječi, bazirano na kontekstu

Riječi koje se nalaze u svim rečenicama imaju manju važnost od riječi koje se javljaju samo u nekim rečenicama

```
# --- LIBRARIES ---
import pandas as pd

from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.naive_bayes import MultinomialNB
from sklearn.model_selection import train_test_split
from sklearn.metrics import accuracy_score, classification_report

# --- PODACI ---
corpus = [
    "burger was good",
    "fries were tasty",
    "service was amazing",
    "burger was not good",
    "fries were not tasty",
```

```
"service was not amazing",
"burger looked good but tasted awful",
"fries looked bad but tasted great",
"great another cold burger",
"amazing service but terrible food"
]

labels = [
    "positive",
    "positive",
    "positive",
    "negative",
    "negative",
    "negative",
    "negative",
    "positive",
    "negative",
    "negative"
]

# --- TRAIN / TEST SPLIT ---
X_train, X_test, y_train, y_test = train_test_split(
    corpus,
    labels,
    test_size=0.3,
    random_state=42
)

# --- TF-IDF ---
vectorizer = TfidfVectorizer(ngram_range=(1,2), stop_words="english")

X_train_tfidf = vectorizer.fit_transform(X_train)
X_test_tfidf = vectorizer.transform(X_test)

# --- MODEL ---
model = MultinomialNB()
model.fit(X_train_tfidf, y_train)

# --- PREDIKCIJE ---
y_pred = model.predict(X_test_tfidf)

# --- REZULTATI ---
print("Accuracy:", accuracy_score(y_test, y_pred))
```

```
print("\nClassification report:\n")
print(classification_report(y_test, y_pred))
```

Accuracy: 0.3333333333333333

Classification report:

	precision	recall	f1-score	support
negative	0.50	0.50	0.50	2
positive	0.00	0.00	0.00	1
accuracy			0.33	3
macro avg	0.25	0.25	0.25	3
weighted avg	0.33	0.33	0.33	3

```
# --- NOVI "TVITОВИ" ZA TEST ---
new_texts = [
    "burger was amazing",
    "burger was not good",
    "fries were not bad",
    "service was terrible",
    "great burger but cold fries",
    "amazing service and tasty food"
]

# --- PRETVARANJE U TF-IDF ---
new_X = vectorizer.transform(new_texts)

# --- PREDIKCIJE ---
predictions = model.predict(new_X)

# --- ISPIS ---
for text, pred in zip(new_texts, predictions):
    print("TEXT:", text)
    print("PREDICTION:", pred)
    print("-" * 40)
```

TEXT: burger was amazing
PREDICTION: negative

TEXT: burger was not good
PREDICTION: negative

```
TEXT: fries were not bad
PREDICTION: negative
-----
TEXT: service was terrible
PREDICTION: negative
-----
TEXT: great burger but cold fries
PREDICTION: negative
-----
TEXT: amazing service and tasty food
PREDICTION: negative
-----
```

```
# --- LIBRARIES ---
from sklearn.feature_extraction.text import CountVectorizer, TfidfVectorizer
from sklearn.naive_bayes import MultinomialNB

# --- PODACI ---
corpus = [
    "burger was not bad",
    "fries were tasty",
    "service was amazing",
    "burger was not good",
    "fries were not tasty",
    "service was not amazing",
    "burger looked good but tasted awful",
    "fries looked bad but tasted great",
    "great another cold burger",
    "amazing service but terrible food"
]

labels = [
    "positive",
    "positive",
    "positive",
    "negative",
    "negative",
    "negative",
    "negative",
    "negative",
    "positive",
    "negative",
    "negative"
]
```

```
# =====
# 1. BAG OF WORDS + NB
# =====
vectorizer_bow = CountVectorizer(ngram_range=(1,2))
X_bow = vectorizer_bow.fit_transform(corpus)

model_bow = MultinomialNB()
model_bow.fit(X_bow, labels)

# =====
# 2. TF-IDF + NB
# =====
vectorizer_tfidf = TfidfVectorizer(ngram_range=(1,2))
X_tfidf = vectorizer_tfidf.fit_transform(corpus)

model_tfidf = MultinomialNB()
model_tfidf.fit(X_tfidf, labels)

# =====
# 3. NOVI TEST PRIMJERI
# =====
new_texts = [
    "burger was amazing",
    "burger was not good",
    "fries were not bad",
    "service was terrible",
    "great burger but cold fries",
    "amazing service and tasty food"
]

# BoW predikcije
new_X_bow = vectorizer_bow.transform(new_texts)
pred_bow = model_bow.predict(new_X_bow)

# TF-IDF predikcije
new_X_tfidf = vectorizer_tfidf.transform(new_texts)
pred_tfidf = model_tfidf.predict(new_X_tfidf)

# =====
# 4. USPOREDBA
# =====
for text, bow_pred, tfidf_pred in zip(new_texts, pred_bow, pred_tfidf):
    print("TEXT:", text)
```

```
print("BoW prediction:  ", bow_pred)
print("TF-IDF prediction:", tfidf_pred)
print("-" * 50)
```

```
TEXT: burger was amazing
BoW prediction:  positive
TF-IDF prediction: negative
-----
```

```
TEXT: burger was not good
BoW prediction:  negative
TF-IDF prediction: negative
-----
```

```
TEXT: fries were not bad
BoW prediction:  positive
TF-IDF prediction: positive
-----
```

```
TEXT: service was terrible
BoW prediction:  negative
TF-IDF prediction: negative
-----
```

```
TEXT: great burger but cold fries
BoW prediction:  negative
TF-IDF prediction: negative
-----
```

```
TEXT: amazing service and tasty food
BoW prediction:  negative
TF-IDF prediction: negative
-----
```

✓ reprezentacija teksta utječe na kvalitetu modela - trebamo bolji traning set

Počnite pisati kôd ili generirati uz pomoć umjetne inteligencije.

